A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 20

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) **1** To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Sea current velocity and direction **1 Properties of the radionuclides:** half-life, solubility, tendency to adsorb onto sediments / bioaccumulate **1** Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people **1** Geology / hydrology of the coast – sediment deposition areas (mud flats) **1 Human activities in the area:** fishing, seaweed collection, beach use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) **1** Monitor their exposure / the concentration of radioactive material in the foods they eat or their body **1** If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk – if not, emissions are controlled **1** CGM assesses exposure before health effects occur (not used to detect damage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Critical Group** → the people most at risk / most heavily exposed	3

Do not write outside the box

Question	Marking guidance	Marks
01.4	**Any 2 of 7** 1 mark each for any 2 of the 7 points below, in any order, maximum 2. **1** Seawater **1** Sediment / mud **1** Seaweed **1** Shellfish / molluscs / fish **1** Milk, vegetables or meat from nearby farms **1** Atmospheric dust **1** Soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** The void fills with water naturally – wetlands are uncommon / valuable habitats in most regions **1** The new habitat may have greater wildlife value than the farmland lost – low cost compared with filling and levelling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.2	**1 mark each** 1 mark for each of the 2 points below. **1 Housing:** ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) **1 Farmland:** landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	1 mark for the point below. **1** Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost – transplanted plants often fail **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 10^3 / 6^3 = 1000 / 216 **1** = 4.6 times (accept 4.63) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
03.2	**1 mark each** 1 mark for each of the 5 points below. **1** C has the highest velocity (most energy, and energy rises with the cube of velocity), strong reliable winds over water, few land-use conflicts, but long expensive cable and offshore installation / maintenance costs **1** A is windy but on a migration route (bird strikes) and peat bog (foundations / tracks damage hydrology; carbon release) – far from grid – scenic objections **1** B is close to the grid with good wind but near a village (noise, visual objection, shadow flicker) - possible with careful siting **1** D is cheap to connect and land can still be farmed between turbines, but low velocity gives little energy **1** Judgement, eg C or B – turbines spaced 3-5 blade diameters apart to avoid wind shadow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	5

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Driven by wind from any direction so no motor to turn them – suitable for turbulent urban sites **1** Generator can be at ground level so a lighter tower **1** Quieter **1** Operate at lower wind velocities **1** Blades more visible so fewer bird deaths **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	2
03.4	1 mark for the point below. **1 No gearbox:** fewer breakdowns, less friction so start generating at lower wind speeds, quieter, lighter nacelle (DC converted to AC by an inverter) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 206-209; AQA spec 3.3.4.1.	
04.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Image databases:** identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features **2** eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance – contamination / DNA transport by water **2 Geolocator tags:** tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate – animal must be recaptured to download data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) **1** Microwaves pass through cloud so monitoring is not interrupted by weather **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1 Droppings:** diet, gender, territory size **1 Owl pellets:** diet **1** Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Aerobic decomposition produces little methane and returns nutrients to the soil	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** It may contain concentrated heavy metals / toxic residues **1** It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
05.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Impermeable polymer / clay liner to stop leachate escaping **1** Collection and treatment of leachate **1** Collection of methane and use as fuel **1** Daily covering of waste / final capping and restoration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	

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Question	Marking guidance	Marks
06.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Single:** 5.8 x 20 x 12 = 1392 W **1 Argon:** 1.6 x 20 x 12 = 384 W **1** Reduction = 1008 W (accept 1010) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The converse of any point, ie the same comparison written the other way round. • The value alone - the unit (W) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** A wider gap of trapped gas reduces conduction because gas is a poor conductor **1** If the gap is too wide convection currents start in the gas, carrying heat across – the optimum depends on the gas and window size **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Low-e coating reflects long-wavelength infrared back into the room, reducing radiation losses **1** Argon / krypton / xenon have lower thermal conductivity than air so less conduction across the gap **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Panes must be held apart by small spacers which cannot be long, so the gap is only about 0.2 mm **1** Used where there is not room for thick gas-filled multiple glazing (eg historic buildings) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.5	1 mark for the point below. **1** Each extra layer saves less energy (diminishing returns) while cost rises, so beyond the optimum the cost of insulation exceeds the value of energy saved **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-231; AQA spec 3.3.4.2.2.	
07.1	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Discharge of tank-washing water from tankers **1** Oil rig / drilling accidents / leakage during drilling **1** Pipeline leaks **1** Leakage from storage tanks / refinery spills **1** Waste lubricating oil from vehicles / industry washed into drains **1** Other ship accidents (fuel oil) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia **1** Feathers lose buoyancy / waterproofing so heavily oiled birds drown **1** Birds swallow toxic oil when preening – time spent cleaning feathers reduces time for feeding themselves / their young **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen **1** Toxic components (benzene, xylene) dissolve and poison plankton near the surface **1** Oil blocks light so algae / phytoplankton cannot photosynthesise **1** Oil masks odours / narcotises so animals cannot find mates or food by smell **1** Shore organisms (molluscs, crustaceans) are covered and asphyxiated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

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Question	Marking guidance	Marks
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Dispersal / dilution by currents and waves is prevented so oil stays concentrated – in open water most spills disperse in days or weeks **1** Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Ozone is very low overnight / early morning, rises from about 08:00, peaks at about 80 units at 14:00 and falls in the evening **1** Ozone is a secondary pollutant formed by photochemical reactions requiring sunlight (UV), so it peaks in the afternoon after several hours of sunlight **1** NO2 (from morning traffic) is broken down by UV to NO and monatomic oxygen – monatomic oxygen reacts with O2 to form ozone **1** Ozone lags behind the NO / NO2 peaks (which are produced by the morning rush-hour traffic) because the reactions take time – falls at night when there is no sunlight **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** NO is the primary pollutant produced in engines when nitrogen and oxygen react at high temperature / pressure **1** NO is oxidised in the atmosphere (by oxygen / ozone) to NO2, a secondary pollutant **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** PANs (peroxyacetyl nitrates) **1** Eye irritation / breathing difficulty / asthma / emphysema / increased risk of heart attacks (toxic at much lower concentrations than NOx) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
08.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Heavy vehicle use giving high NOx and hydrocarbon emissions **1** Sunny climate providing the UV for photochemical reactions **1** Valley / basin topography and frequent temperature inversions trapping pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 248-249, 253-254; AQA spec 3.4.3.2.3-3.4.3.2.4.	
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury **1** Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes **1** It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water **1** Local people ate large amounts of fish (critical group) so received toxic doses **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **critical group** → the people most at risk / most heavily exposed **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) **1** It crosses the placenta and harms the developing unborn baby **1** Mercury is a teratogen, preventing normal gene expression / development **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 3 points below. **1** Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs **1** Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut **1** Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs – liposoluble so most toxic and bioaccumulates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.	
10.1	**1 mark each** 1 mark for each of the 3 points below. **1** Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) **1** They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone **1** The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	3

Do not write outside the box

Question	Marking guidance	Marks
10.2	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Provides carbon, oxygen, hydrogen and nitrogen (as N_2, O_2, CO_2, H_2O) for biological molecules **1** Delays the escape of infrared energy, keeping the surface warm **1** Distributes heat and water vapour by winds / ocean currents **1** Provides atmospheric pressure so liquid water can exist **1** Deflects damaging radiation of the solar wind in the upper atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
10.3	1 mark for the point below. **1** Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **stratified** → sampling each area or group in proportion to its size	1

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Troposphere:** enhanced greenhouse effect - anthropogenic CO_2 (fossil fuels, deforestation), methane (rice, livestock, landfill), N_2O, CFCs, tropospheric ozone – more infrared absorbed and re-emitted – aerosols and smoke increase albedo / scatter light – land-use change alters surface albedo and evapotranspiration • **Stratosphere:** CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium – less UV absorbed in the stratosphere, more reaches the surface – stratospheric cooling • **Consequences:** changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions – increased UV damage to organisms • **Difficulties of prediction:** linked processes where one change alters others – feedback (positive and negative) and tipping points – time lags (ocean heat capacity, long residence times of CFCs and CO_2) – natural variability (solar output, volcanoes, Milankovitch cycles) – incomplete understanding of atmospheric processes – limited historical data so proxies needed – models with different assumptions • **Evaluation:** some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain – precautionary principle • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **enhanced greenhouse effect** → extra warming caused by added greenhouse gases • **anthropogenic** → human-caused / caused by human activity • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	In the troposphere human activity enhances the greenhouse effect: fossil fuels and deforestation add CO_2, rice, livestock and landfill add methane, fertilisers add nitrous oxide, and CFCs and tropospheric ozone are new absorbers. More outgoing infrared is absorbed and re-emitted, warming the surface, while smoke and aerosols scatter sunlight and land-use change alters albedo and evapotranspiration. In the stratosphere, CFCs, halons and HCFCs release chlorine and bromine that catalyse ozone destruction, shifting the O / O_2 / O_3 equilibrium so that less UV is absorbed there and more reaches the surface, cooling the stratosphere as well.	
	The consequences run through every system the atmosphere's energy drives: climate, thermohaline circulation, the hydrological cycle, sea level and species distributions, plus UV damage to organisms. Predicting them is hard because the processes are linked - one change alters others - and because feedback can amplify (albedo loss, methane release) or damp (cloud cover, photosynthesis) the effect, sometimes past tipping points. Time lags from ocean heat capacity and the long residence times of CO_2 and CFCs, natural variability from solar output, volcanoes and orbital cycles, sparse historical data replaced by proxies, and models built on different assumptions all add uncertainty.	
	Some things are nevertheless well established: the physics of infrared absorption, the chemistry of ozone loss, and the success of the Montreal Protocol. What is uncertain is the scale and timing of regional impacts, which is exactly why the precautionary principle, not certainty, should guide action.	
	Weaker answer - Level 1 (3 marks)	
	Humans release greenhouse gases like carbon dioxide from burning fossil fuels into the troposphere, which traps more heat and causes global warming. CFCs go into the stratosphere and destroy the ozone layer, so more UV reaches the Earth. It is hard to predict what will happen because the atmosphere is complicated and scientists do not know everything about it. There could be feedback effects that make it worse.	
	What makes the difference	
	• Level 1 gives one cause per layer; Level 3 lists several gases and processes and explains the mechanism (absorption and re-emission; catalytic destruction shifting an equilibrium).	
	• 'Difficulties of prediction' is answered in Level 3 with specific reasons (linked processes, feedback, time lags, natural variability, proxy data, models); Level 1 says 'complicated'.	
	• Level 3 evaluates - separating what is certain from what is not - and links this to the precautionary principle.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Scale of the problem:** how energy is used (transport, buildings, industry, water) – links between affluence and per capita use • **Transport:** bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • **Buildings:** orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • **Industry:** heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • **Limits:** rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples – comparison with the cost of new supply	25
	– conclusion that conservation is the cheapest first step but cannot alone meet demand growth *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Composition and structure:** N_2, O_2, Ar, CO_2, ozone – troposphere and stratosphere – pressure and liquid water – gases for biological molecules and industry • **Energy processes:** UV absorbed by O / O_2 / O_3 dynamic equilibrium – visible light absorbed at the surface – infrared absorbed by greenhouse gases (33 °C warming) – heat distribution by winds and ocean currents – transport of water vapour – energy budget and albedo • **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species – difficulties of monitoring and prediction – control strategies and agreements • **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • **Other threats:** aerosols, smoke, land-use change – interconnection of processes	25
	• **Evaluation:** which functions are most at risk – success of ozone controls vs slow progress on greenhouse gases – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	

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